

6 The human body responds to changes in both its internal and external environments.

(a) How many of these statements describe the effects of adrenaline in the “fight or flight” response?

- arterioles constrict in skeletal muscles
- glycogen is released from the liver
- the rate of food digestion decreases

(1)

☐ **A** 0

☐ **B** 1

☐ **C** 2

☐ **D** 3

(b) The heart responds to adrenaline by increasing cardiac output.

Cardiac output = heart rate $\times$ stroke volume

(i) Calculate the cardiac output for a person who has a heart rate of 77 bpm and a stroke volume of 70 cm^3 .

Give your answer to two significant figures in $\text{dm}^3 \text{ hour}^{-1}$.

(2)

Answer $\text{dm}^3 \text{ hour}^{-1}$

- (ii) The formula shows how to calculate the cardiac index. This is a method of adjusting cardiac output values for the body surface area of an individual.

$$\text{cardiac index} = \frac{\text{cardiac output in dm}^3 \text{ min}^{-1}}{\text{body surface area in m}^2}$$

The table shows some information for cardiac output, body surface area and cardiac index for two individuals.

Individual	Cardiac output / dm ³ min ⁻¹	Body surface area / m ²	Cardiac index
A	6.7	1.92	
B		1.49	2.8

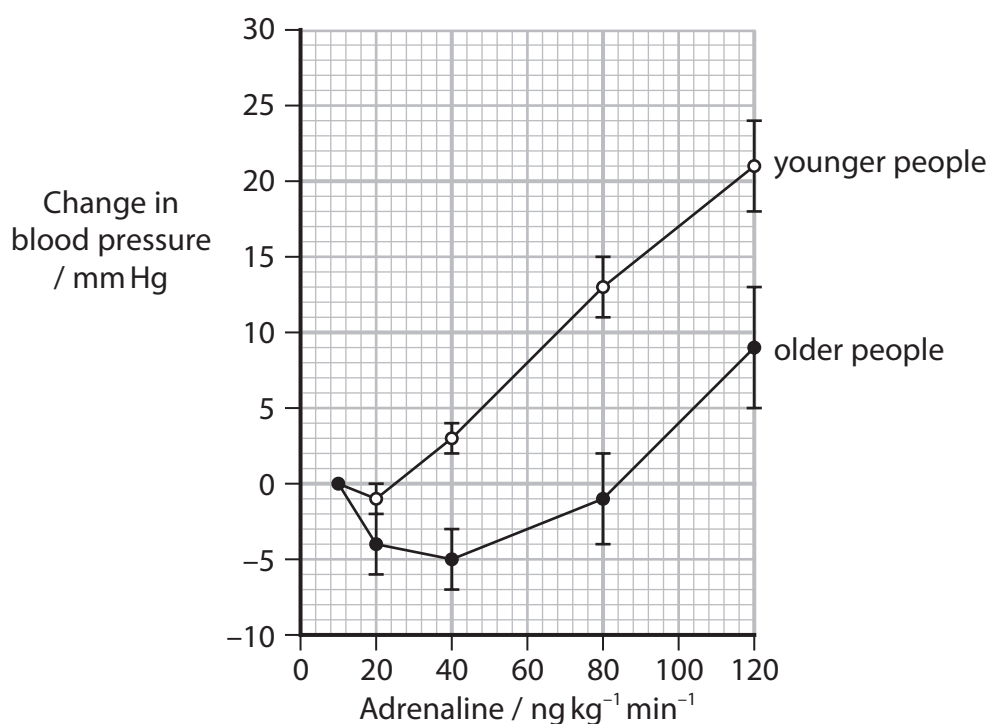
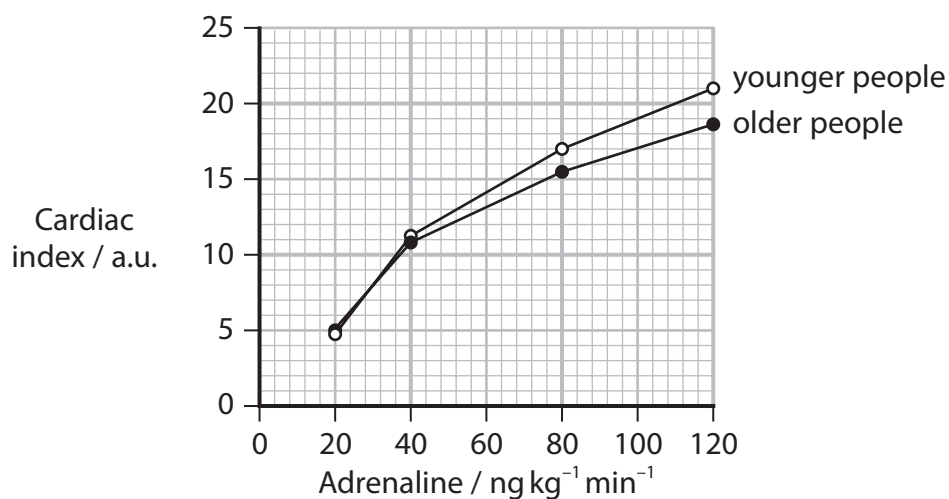
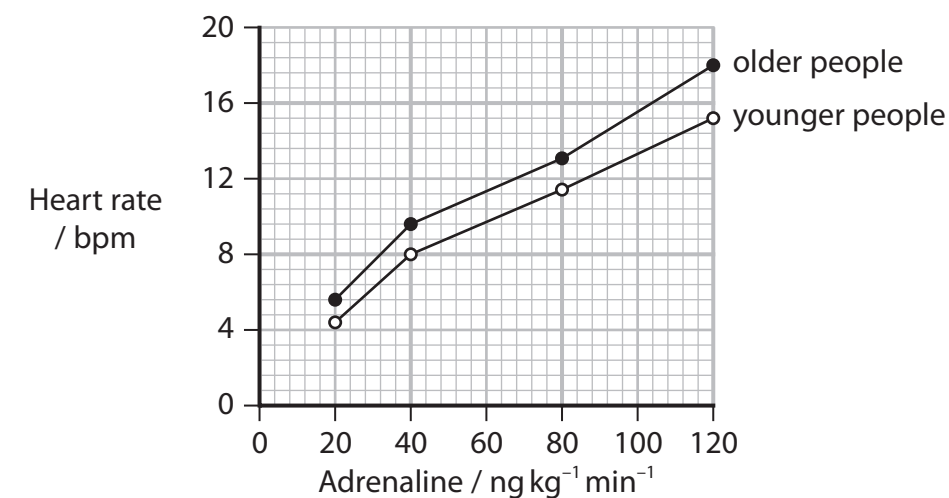
Complete the table to show the units for cardiac index, the cardiac index for individual A and the cardiac output for individual B.

(3)

*(iii) Ageing is associated with changes in the way the heart responds to adrenaline.

In an investigation, the effects of adrenaline on the hearts of young people and old people were recorded.

The graphs show the results of this investigation.



Discuss the effect of ageing on the response of the heart to adrenaline.
Use the information in the graphs to support your answer.

(6)

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(Total for Question 6 = 12 marks)